

KRISHNA MADANI

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EDUCATION

Northeastern University

Master of Science in Information Systems

Boston, MA

Sep 2023 – Dec 2025

University of Mumbai

Bachelor of Engineering in Information Technology

Mumbai, India

Aug 2016 – May 2020

EXPERIENCE

Crewasis.ai

Software Engineer

New York City, NY

Feb 2025 – Jun 2025

- Reduced payload and API response overhead in service-oriented **React.js** web services by refactoring data-fetching boundaries and introducing route-level code splitting, cutting page load time by **40%** across all production deployments
- Built a multi-format file upload and download UI in **React.js**, backed by server-side presigned **S3** URL generation with file type, size and expiration validation via **AWS IAM** policies, enforcing authentication and authorization policies
- Containerized the **React.js** frontend and **Django** backend using **Docker** and pushed images to **AWS ECR**, enabling consistent and repeatable CI/CD deployments across staging and production via **AWS ECS**
- Implemented client-side debounce on a **React.js** search interface backed by prefix-optimized **PostgreSQL** queries and response caching, cutting backend request volume by **90%** while sustaining **sub-500ms** API response times

Capgemini

Software Engineer

Mumbai, India

Sep 2020 – Jul 2023

- Engineered an event-driven serverless gift card system in **Node.js** as a service-oriented system on **AWS Lambda, DynamoDB, S3, SQS** and **API Gateway**, achieving a **99%** transaction success rate across 200 franchise locations
- Implemented a **GraphQL** API layer replacing fragmented **REST** web services, enabling flexible and scalable data queries that cut average response times from **2s to 450ms** across production services
- Designed low-level system design for **RESTful** microservices architecture using **OOP** principles and **OpenAPI** during a SOAP-to-REST migration, delivering a distributed API layer that improved Loyalty platform response times by **30%**
- Optimized DynamoDB schema and access patterns using **GSI/LSI indexing** strategies and introduced **DAX** as an in-memory cache layer, reducing data-access latency by **40%** across high-traffic query APIs and enhancing cache performance
- Streamlined deployment pipelines via **Azure DevOps, CloudFormation** and **Serverless Framework** in an Agile/Scrum SDLC, gating every PR on test coverage and build validation while aligning delivery with cross-functional stakeholders

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, SQL

Backend: Node.js, Express.js, Spring Boot, Django, Flask, FastAPI

Frontend: React.js, Next.js, HTML, CSS/SASS, Vite, Redux, WebSockets

Cloud and DevOps: AWS, Azure DevOps, Docker, Kubernetes, CI/CD, Serverless Framework

Databases and Tools: PostgreSQL, MongoDB, MySQL, Git, GitHub Actions, Jenkins, OpenAPI, Postman, Jira, SonarQube

PROJECTS

WordArena | React, Node.js, Express.js, Socket.io, MongoDB

- Architected a real-time multiplayer platform with **WebSocket**-based server-side state isolation to prevent client data exposure and automated role-swapping between setter and guesser across rounds
- Replaced hardcoded word lists with a **MongoDB**-backed word bank, using category-based randomization and a progressive hint system that scales disclosure by wrong-guess count to deliver adaptive difficulty without manual configuration

ContentFlow | Python, CrewAI, Flask, ChromaDB

- Built a 5-agent AI orchestration pipeline with **ChromaDB**-backed RAG for shared memory, automated 0–100 quality scoring, and threshold-based retry logic enforcing output standards
- Engineered dual-tier **LLM** failover (Gemini → Groq Llama) validated through real rate-limit failures, delivering publication-ready articles in **~5 minutes** via 3 custom NLP tools built on NLTK and TextBlob

AuctionHub | Java, Spring Boot, PostgreSQL, WebSocket, Vite

- Delivered a high-throughput bidding engine with tiered minimum increment logic and live bid propagation to all connected clients via **STOMP WebSocket** using `SimpMessagingTemplate`
- Implemented configurable anti-sniping logic that dynamically extends auction end time when bids land within a threshold window, broadcasting real-time extension events to all subscribers